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Investigation and evaluation of building energy flexibility with energy storage system in hot summer and cold winter zones

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ABSTRACT

The dramatically increasing energy demand of building air-conditioning in hot summer and cold winter zones fluctuates greatly, especially during the period of cooling and heating in summer and winter, which exerts strict operation burden and challenge on power-grid system stability. In order to realize building energy flexibility, this paper proposes a type of energy utilization system including electricity chiller, gas-boiler, electricity heater, cold and thermal energy storage tank (case II), on the basis of electricity chiller and gas-boiler for cooling and heating demand (case I), which is commonly used as cold and heat source in hot summer and cold winter zones. The total cost of the whole life period and annual carbon emission of energy systems are served as objective function is optimized. Furthermore, the system capacity confirmation and load demand management control are analyzed and discussed, the characteristics of energy-flexibility with and without energy storage are presented, the load coefficient is introduced to evaluate the impact of case II on building energy flexibility. Compared with case I, with the deployment of cold and thermal energy storage, the cost-effectiveness of case II is 16.5% and carbon emission is reduced by 30.9% compared with case I .In case II, the load coefficient of electricity chiller and electricity heater is 83.34% and 30.65% respectively, which means that energy storage system enables to promote building energy flexibility improvement and grid-friendliness enhancement. This paper provides methods and theoretical guidance for the planning-operation of building energy system and building energy flexibility in hot summer and cold winter zones.

1. Introduction

The increasing energy demand, especially the peak power demand, has exerted great operation burden and challenge on the power grid system during peak hours [1,2]. In order to satisfy the peak power demand, the power system must be equipped with power generation equipment with larger installed capacity, which not only increases the system initial investment cost, but also has adverse impact on the flexibility of power production and power grid balance [3]. Therefore, flexible energy consumption is of great significance to power grid. Building energy consumption accounts for about 1/3 of the energy consumption of the whole society, and the building energy consumption is volatile and fluctuates dramatically with time, so building sector has great potential to provide energy flexibility for the power grid [4].

At present, the methods to perform building energy-flexible electricity utilization mainly include peak load shifting control strategy and energy storage technology [5,6]. Peak load shifting control management means that smooth the power supply curve of power grid without changing the total energy consumption, the peak power demand is reduced by employing advanced load management technology, and the flexible load is shifted from peak period to off-peak period [7–9]. Many researches have been conducted on load shifting control strategy by scholars at home and abroad. Jin et al. [10] proposed an integrated optimal scheduling and predictive control method based on building

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thermal mass management considering the thermal dynamics of building envelope with multiple layers of material, and the results shown that compared with the traditional rule-based control method, the proposed energy consumption prediction optimization method can significantly reduce the peak-valley load difference and minimize the operating cost. Gao et al. [11] proposed a building thermal mass energy management strategy based on machine learning, which seeks to satisfy the thermal comfort requirements while minimizing the cost of electricity consumption and reducing the building energy demand during peak hours. Sánchez Ramos et al. [12] analyzed the demand management strategy based on building thermal mass as thermal storage. After the implementation of demand management, flexible building energy consumption can be realized, and the maximum economic benefit saving is equivalent to 3.2% of heating and 8.5% of cooling. Moazenia et al. [13] implemented building energy management on the case building with islanded water-energy microgrid system, they determined the optimal water demand and peak daily operation strategy of the refrigerator. Liu et.al [14] indirectly scheduled the micro-grid and household energy management system to solve the unbalanced power problem between energy generation and demand, which mainly relied on price signals, incentives, and measures to avoid the simultaneous use of equipment (e.g. turning off energy-consuming equipment such as dishwashers, washing machines, and water heaters on the demand side). Yang [15] changed retailers' electricity purchase demand based on energy storage demand response to reduce electricity costs and enhance the reliability of the grid as soon as possible.

Some research on the application of energy storage system in air conditioning system have been conducted. Hassani et al. [16]. presented a novel cold storage system integrated with phase change material plate and condenser in air-conditioning system is studied in order to improve its coefficient of performance. Aljehani et al. [17] studied the phase change composite as a potential storage medium for cold thermal energy storage integrated with the steam compression refrigeration cycle in air conditioning system are studied. The results shown that the system produces significant benefits in terms of compressor size, compressor efficiency, power consumption and carbon dioxide emissions. However, the operation optimization of energy storage system and the additional investment cost of the integration the air conditioning system and cold storage system were not involved. Congedo et al. [18] proposed the application on microscale of an innovative trigeneration system with a micro Compressed Air Energy Storage system and the integration of renewable energy production, focusing on the potential to cover the electrical loads of the air conditioning and domestic hot water demand by maximizing the storage capacity of the system in integrated mode and independent operation mode. Ashok et al. [19] developed a methodology to determine the optimal chilled water storage capacity and corresponding operating strategy for air conditioning loads for different electricity price by nonlinear optimization. Hoseini Rahdar et al. [20] integrated the ice energy storage system with the office building air conditioning system, and non-dominated sorting genetic-algorithm II and particle swarm optimization algorithm were applied to model and study the exergy, economic and environmental factors of the refrigeration system, the optimal design parameters were obtained, and the energy demand of the energy storage system and the traditional system were compared. Cui et al. [21] developed a model-based optimization design method, which provides a cost-saving potential method for the integration of active cold storage and HVAC systems in commercial buildings, and used genetic-algorithms to optimize the capacity of active cold storage systems to maximize life cycle costs. However, their research is only based on the rule operation strategy for capacity optimization, without taking into account the system operation optimization and electricity price peak-valley difference for energy storage and the principle of minimum carbon emissions. Soler et al. [22] developed a reliable and fast algorithm for chillers with thermal storage system for minimizing the operation costs through solving a linear or convex optimization problem. Based on energy and mass balance principles, Ono et al. [23] considered the variable conditions of chilled water temperature and applied the nonlinear programming method to study the operation optimization problem of thermal system consisted of chiller, air-conditioning system and thermal storage tank for reducing the running cost of a day and energy consumption. Zhang et al. [24]. presented a method of determining the optimal tank size of stratified chilled water storage tank as well as corresponding optimal operating strategies for reduce the utility billing cost and payback period. Their research only involved the energy-saving benefits, without considering the optimization of capacity configuration of energy storage system and minimizing the total life cycle cost of energy system.

Most of the previous studies are based on the operation rules to optimize energy storage capacity, and few studies consider to optimize the capacity and operation of the energy storage system at the same time. Furthermore, in previous studies, their research only involved the energy-saving benefits or daily operation cost reduction, without considering the optimization of capacity configuration of energy storage system in the premise of minimizing the total life cycle cost of energy system and carbon emissions by multi-objective optimization. More importantly, building electricity energy flexibility is rarely involved in the previous cold and heat source system optimization research with energy storage system. With the popularity of building electrification, building electricity energy flexibility consumption is a good and fantastic research topic, which is deserved to our attention. Accordingly, there is a research gap in developing the optimal and effective capacity confirmation and operation optimization for the planning and operation of building cold and heat source system whilst minimizing the total life cycle cost and annual carbon emissions of the energy storage system. Therefore, the main goal of this paper is to establish the building cold and heat source system optimization model with the minimum life cycle total cost and carbon emission of energy storage system based on bilevel optimization method. Then, the capacity configuration and operation optimization of building cold and heat source systems are optimized based on bilevel optimization framework, and the load coefficient is introduced to determine and evaluate the building electricity flexibility of two building cold and heat source systems, the load demand management control strategy is obtained.

The innovations of this study can be summarized as follows:

- The capacity optimization and operation optimization in 8760 h for building heating and cooling source system is conducted based on bilevel optimization framework.
- The life cycle total cost reduction and carbon emission reduction is obtained for energy storage system by multi-objective optimization.
- The building electricity flexibility with energy storage system is considered in hot summer and cold winter zones, and load coefficient is introduced to evaluate energy flexibility and the shaving peak demand into valley period is achieved.

2. Optimization model of building energy systems

In view of the characteristics of building energy demand in hot summer and cold winter zones, energy storage system and gas boiler plus electricity chiller (i.e. reference system case I) are employed to provide energy demand for the building, and the optimization model of cold and heat source system in hot summer and cold winter zones is established in this part, which is used to study the adaptability of different cold and heat source systems on building energy flexibility in hot summer and cold winter areas.

2.1. Description of building energy systems

In this paper, two families of heating and cooling systems are designed to study the performance of energy-flexibility of energy storage. As illustrated in Fig. 1(a), gas boiler (GB) and electricity chiller (ECH) are welcomely accepted as the cooling and heating source scheme

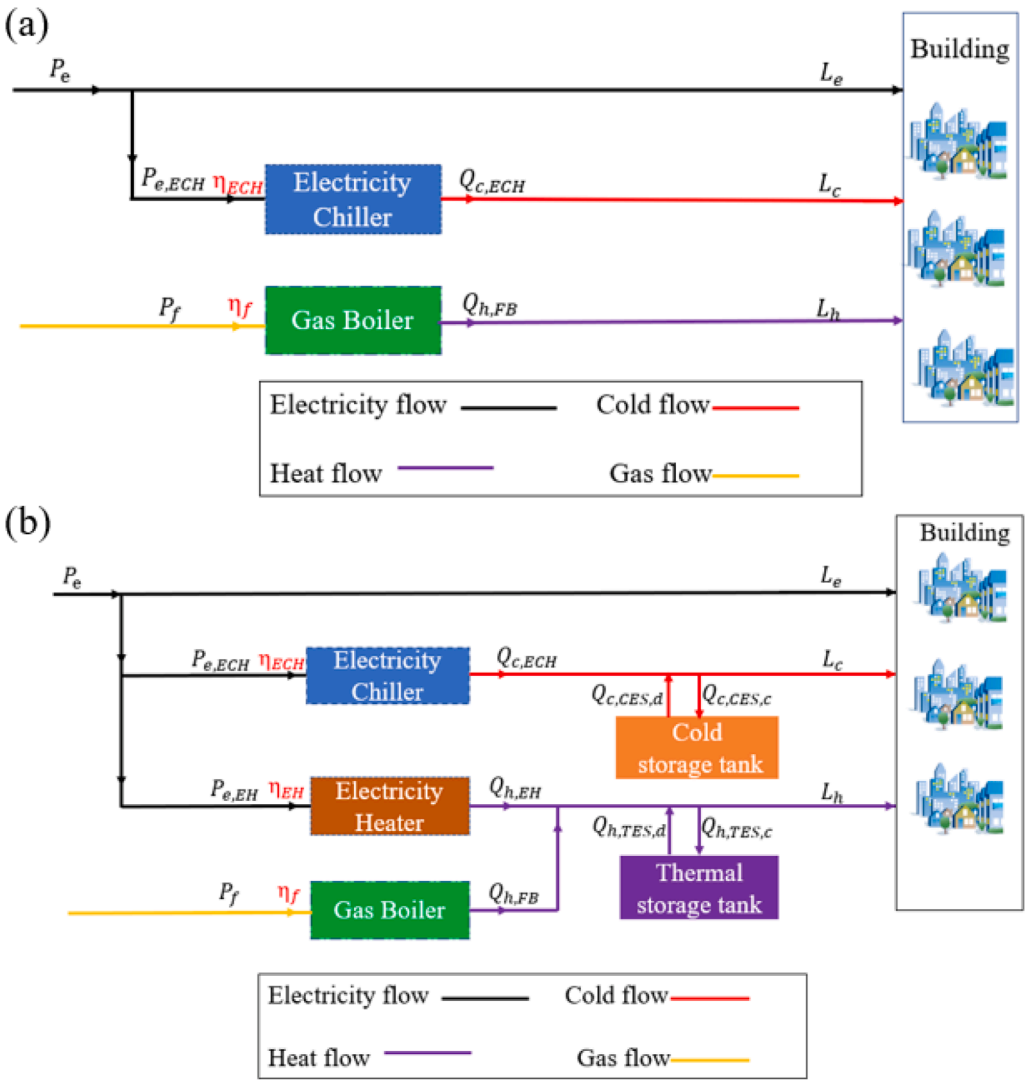


Fig. 1. Diagram of building energy systems.

in hot summer and cold winter zones to provide heat and cold energy respectively for buildings, which is regarded as case I (i.e. electricity chiller plus gas-boiler for cooling and heating demand). For the purpose of mitigating the unfavorable consequence of peak energy demand in summer and winter on power grid and utilization of energy flexibility as well as maintaining power grid balance, this paper intends to employ electricity heater and energy storage systems for energy-supplying based on case I for the purpose of shaving the peak and filling the valley and smoothen the power fluctuation. The new cold and heat source scheme of ECH, electricity heater (EH), GB, cold storage water tank (CES) and thermal storage tank (TES) is regarded as case II. As shown in Fig. 1(b), in the new scheme of case II, the cold load is shared by ECH and CES, and the heat load is satisfied by EH, GB, and TES. On account of the introduction of cold and thermal storage tank, the building energy system can store cold and heat energy during low electricity prices at night, so as to shift peak demand into valley period and reduce the shock and disruption of electricity load on the grid during peak hours.

The Hongchuang Times Square is considered as the research project in Changsha city. The physical performances of this building are as follows: 5 floors above the ground, 2 floors below the ground, building height of 23.6 m, overall construction area of 16,100 m^2s, mainly served as shopping malls and hotels. The annual hourly cold and heat load of this park is depicted in Fig. 2. It can be seen from the figure that the heat load of this building is mainly concentrated in January-March and November-December, and the peak heat load is 271.24 kW. The building cold load mainly distributed from May to August, and the peak cold load is 803.24 kW.

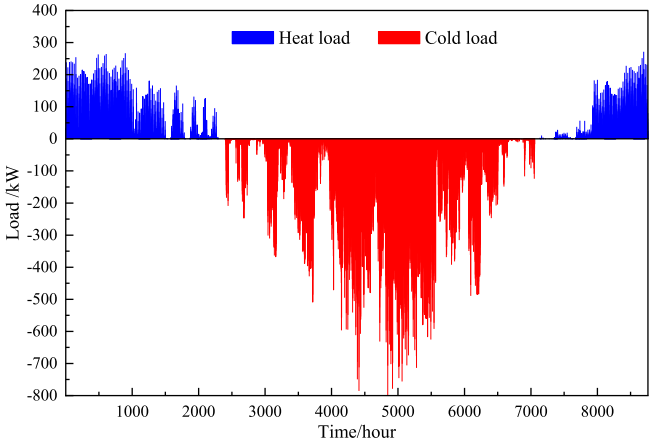


Fig. 2. Annual air-conditioning load in Hongchuang Times Square.

2.2. System modeling

2.2.1. Equation of electricity energy balance

According to the electricity flow in Fig. 1(b), the electricity demand of the building was composed of three parts: the power consumption of electricity chiller ($P_{e,ECH}(t)$), electric heater ($P_{e,EH}(t)$), lighting load and other electricity demand (which is symbolic of L_e). Therefore, the total electricity power demand $P_e(t)$ can be wrote by Eq. (1):

$$P_{e,ECH}(t) + P_{e,EH}(t) + L_e = P_e(t) \tag{1}$$

The units of $P_{e,ECH}(t)$, $P_{e,EH}(t)$,L_e all are kW. It is noteworthy that the whole lighting and other electricity demand L_e is constant, and the value of it is 65,000 kW for around year, equivalent to 7.42 kW per hour.

2.2.2. Thermal energy balance equation

The thermal demand of the case building derives from gas boiler (GB), electricity heater (EH) and thermal storage system as descripted in Fig. 1(b).Therefore, the thermal balance equation of the system can be shown as Eq. (2):

$$Q_{h,EH}(t) + Q_{h,FB}(t) + Q_{h,TES,d}(t) = L_h(t) + Q_{h,TES,c}(t) \tag{2}$$

Where,$Q_{h,EH}(t)$ and $Q_{h,FB}(t)$ refer to the heating quantity of the electricity heater and gas boiler respectively(kW). The subscript h, d, c means heating, discharging energy, and charging energy respectively; $L_h(t)$ is the thermal load of the building(kW).

2.2.3. Cold energy balance equation

The cold load is borne by the electricity chiller and the cold storage water tank, so the cold energy balance equation can be formulated as Eq. (3):

$$Q_{c,ECH}(t) + Q_{c,CES,d}(t) = L_c(t) + Q_{c,CES,c}(t) \tag{3}$$

where, $Q_{c,ECH}$(t) is the cooling capacity of the electricity chiller (ECH) and the subscript c symbolizes cold energy; $Q_{c,CES,d}$(t) and $Q_{c,CES,c}(t)$ refer to the released cooling capacity and charging cooling capacity of the cold storage tank at time t respectively, and the subscript d and c are cold discharging and the cold charging respectively; L_c(t) is the building cold load.

2.2.4. Energy storage system balance

The stored capacities of the energy storage tanks are the function of the charging and discharging energy at this moment well as the stored energy at the last moment. In addition, the energy of these energy storage systems will dissipate with time. Therefore, the energy storage system balance can be set as Eq. (4) and (5):

$$Q_{h,TES}(t) = \eta_{TES}.Q_{h,TES}(t-1) + \eta_{TES,c}Q_{h,TES,c}(t) - \frac{1}{\eta_{TES,d}}Q_{h,TES,d}(t) \tag{4}$$

Where,$Q_{h,TES}(t)$ is the net stored heat at time t in thermal storage tank; η_{TES} is the heat storage efficiency of the heat storage tank, which is 0.95;$Q_{h,TES}(t-1)$ is the net stored heat of the storage tank at time (t-1);$\eta_{TES,d}$ and $\eta_{TES,c}$ are the heat discharging efficiency and heat charging efficiency of the heat storage tank, both of which are 0.95.

The characteristics of the cold storage tank and heat storage tank are similar, and both have energy dissipation. The cold storage system model satisfies Eq. (5):

$$Q_{c,CES}(t) = \eta_{CES}.Q_{c,CES}(t-1) + \eta_{CES,c}Q_{c,CES,c}(t) - \frac{1}{\eta_{CES,d}}Q_{c,CES,d}(t) \tag{5}$$

Where, $Q_{c,CES}(t)$ and $Q_{c,CES}(t-1)$ are the net stored cooling capacity of cold storage tank at time t and time (t-1) respectively; η_{CES} is the storage efficiency of the cold storage tank (η_{CES}=0.92); $\eta_{CES,d}$ and $\eta_{CES,c}$are the cold discharging efficiency and cold charging efficiency of the cold storage tank respectively, both of which are 0.92.

2.2.5. Energy equipment constraints

In order to ensure that the energy-supplying system can provide the energy demand of the building, each energy carrier in the energy system should comply with the corresponding equipment constraints all period. Among them, the electricity chiller should meet Eq. (6) - (7):

$$Q_{c,ECH}(t) = \eta_{ECH}.P_{ECH}(t) \tag{6}$$

$$Q_{c,ECH}(t) \le Q_{c,ECH,max} \tag{7}$$

Where, η_{ECH} is the cooling efficiency of electricity chiller (η_{ECH}=4.0);$Q_{c,ECH}(t)$ is the cooling capacity of the electricity chiller and $P_{ECH}(t)$ is the electricity power consumed by the electricity chiller; $Q_{c,ECH,max}$ is the maximum cooling capacity of the electricity chiller, also taken as the optimal cooling capacity (kW) in this research.

Similarly, gas boiler corresponds to (8)—(9) constraints:

$$Q_{h,FB}(t) = \eta_f.P_f(t) \tag{8}$$

$$Q_{h,FB}(t) \le Q_{h,FB,max} \tag{9}$$

η_f is the heating efficiency of the gas boiler, which is 0.85;$P_f(t)$ is the natural gas power consumed by the gas boiler (kW). $Q_{h,FB,max}$ is the maximum heating capacity of the gas boiler, also taken as the optimal heating capacity of the gas boiler (kW).

In the same way, the constraints of electricity heater are listed as Eq. (10)-(11):

$$Q_{h,EH}(t) = \eta_{EH}.P_{e,EH}(t) \tag{10}$$

$$Q_{h,EH}(t) \le Q_{h,EH,max} \tag{11}$$

Where, η_{EH} is the heating efficiency of the electricity heater, which is 0.95; $Q_{h,EH}(t)$ is heating capacity of the electricity heater at time t (kW); $P_{e,EH}(t)$ is the electricity power consumed by the electricity boiler at time t (kW); $Q_{h,EH,max}$ is the maximum heating capacity of the electricity boiler, also regarded as the optimal heating capacity of the electricity heater (kW).

The heat and cold energy storage capacity constraints shown in Eq. (12)-(14) and Eq. (15)-(17) respectively:

$$0 \le Q_{h,TES,c}(t) \le Q^{max}_{h,TES,c} \tag{12}$$

$$0 \le Q_{h,TES,d}(t) \le Q^{max}_{h,TES,d} \tag{13}$$

$$0 \le Q_{h,TES}(t) \le Q^{max}_{h,TES} \tag{14}$$

$$0 \le Q_{c,CES,c}(t) \le Q^{max}_{c,CES,c} \tag{15}$$

$$0 \le Q_{c,CES,d}(t) \le Q^{max}_{c,CES,d} \tag{16}$$

$$0 \le Q_{c,CES}(t) \le Q^{max}_{c,CES} \tag{17}$$

Where, $Q^{max}_{h,TES,d}$and $Q^{max}_{h,TES,c}$ are the maximum heat discharging capacity and the maximum heat charging capacity of the thermal storage tank respectively, and $Q^{max}_{h,TES}$ is the maximum heat storage capacity of the heat storage tank, also can be known as the optimal capacity in this study. In terms of cold energy storage system, $Q^{max}_{c,CES,d}$ and $Q^{max}_{c,CES,c}$ are maximum cold discharging capacity as well as the maximum cold charging capacity; and $Q^{max}_{c,CES}$ is the maximum cooling storage capacity of the cold storage system, also taken as the optimal cooling capacity of the cold storage system.

This research introduces energy storage tanks to store energy, so it is necessary to determine reasonable and feasible storage tank volume. The calculation of energy storage tank volume uses formula (18):

$$V_{yES}(t) = \frac{3.6 \times Q_{x,yES}(t).K_1K_2H}{\rho c_p(T_{max} - T_{min})} \tag{18}$$

Where, $V_{yES}(t)$ is the volume of the energy storage tank at time t (m^3); x can be either h or c, depending on whether heating or cooling is considered; yES can be either thermal storage system (TES) or cold storage system (CES); K_1 and K_2 are the safety factors considering the heat loss or cold loss caused by the delivery process (K_1=K_2=1.10); ρ is the density of water (ρ= 1000 kg/m^3); c_p is the specific heat of water (c_p=4.21$kJ/kg{\cdot}K$); T_{max} is the average maximum temperature of the water tank when the energy storage process is finished; T_{min} is the minimum average temperature of water in the storage tank at the beginning of energy storage. The temperature difference between T_{max} and T_{min} is 45 °C . H is energy storage time during the valley period (H = 8).

The constraint of the energy storage tank is shown in Eq. (19).

$$0 \le V_{yES}(t) \le V_{yES}^{max} \tag{19}$$

Where, V_{yES}^{max} is the maximum volume of the energy storage tank (m^3).

2.3. Objective function

The total cost of the energy system is a primary concern in engineering practice. In addition, carbon emission is also an important indicator of system evaluation. Therefore, for the energy storage system (i.e.case II) proposed in this paper for energy flexibility consumption in buildings, the lowest carbon emission and the minimum total cost in the whole life cycle are selected as the goals, so the optimization problem studied in this paper can be expressed as Eq. (20):

$$min\left(\omega C_{total} + (1-\omega) CO_{2,\ total}\right) \tag{20}$$

Where, ω is the weight given to the minimum total cost (C_{total}) of the first objective function, the value of ω is varied from 0 to 1. And $CO_{2,\ total}$ is the minimum carbon emission of the second objective function.

The calculation of the total cost$C_{total}$is followed as Eq. (21):

$$C_{total} = C_{init} + C_{op}{\cdot}\frac{(1+i)^n - 1}{i(1+i)^n} \tag{21}$$

Where, i is the annual interest rate (i = 5%); n is the payback period, which is 10 years; C_{init} and C_{op} are the initial capital and operational cost of the system. The initial capital is calculated by Eq. (22).

$$C_{init} = \varphi_{FB}{\cdot}Q_{h,FB,max} + \varphi_{CH}{\cdot}Q_{c,ECH,max} + \varphi_{EH}{\cdot}Q_{h,EH,max} + \varphi_{TES}{\cdot}V_{TES}^{max} + \varphi_{CES}{\cdot}V_{CES}^{max} \tag{22}$$

Where, φ_{FB} is the price of the unit capacity of the gas boiler (φ_{FB}=166 RMB/ kW);$Q_{h,FB,max}$ is the maximum heating capacity of the gas boiler; φ_{CH} is the price of the unit capacity of the electricity chiller (φ_{CH}=355 RMB/kW); $Q_{c,ECH,max}$ is the maximum cooling capacity of the electricity chiller; φ_{EH} is the price of the unit capacity of the electricity heater (φ_{EH}=195 RMB/kW); $Q_{h,EH,max}$ is the maximum heating capacity of the electricity heater; φ_{TES} is the price of the unit volume of the heat storage tank (φ_{TES}=320 RMB / m^3); V_{TES}^{max} is the maximum volume of the heat storage tank; φ_{CES} is the price of the unit volume of the cold storage tank (φ_{CES}=320 RMB/m^3), V_{CES}^{max} is the maximum volume of the cold storage tank.

The operating cost (C_{op}) is separated into two parts: electricity energy operating cost and gas-fire operating cost. The operation cost is calculated according to Eq. (23):

$$C_{op} = C_{f,FB} + C_{e,op} \tag{23}$$

The operation cost of gas-fire ($C_{f,FB}$) can be calculated by Eq. (24).

$$C_{f,FB} = \varphi_f \sum_{i=1}^{N} P_f(t).\Delta t \tag{24}$$

Where, φ_f is the unit price of natural gas, which is 3.28 RMB/m^3(0.328RMB/kW•h), and $P_f(t)$ is the power consumed by the gas boiler at time t (kW); Δt is the time step (h), and N is the annual operation time (N = 8760 h).

The operation cost of electricity energy ($C_{e,op}$) is calculated by Eq. (25).

$$C_{e,op} = \begin{cases} \varphi_{e,f}{\cdot}\sum^{P_e}(t).\Delta t & t \in T_f \\ \varphi_{e,p}{\cdot}\sum^{P_e}(t).\Delta t & t \in T_p \\ \varphi_{e,g}{\cdot}\sum^{P_e}(t).\Delta t & t \in T_g \end{cases} \tag{25}$$

$\varphi_{e,f}$ is the electricity price of peak period; $\varphi_{e,p}$ is the electricity price of the flat period; $\varphi_{e,g}$is the electricity price of the valley period. The time-of-use tariff is shown in Table 1. T_f is the operation time of the whole year during peak period; T_p is the operation time of the whole year during the flat period; T_g is the operation time of the whole year during the valley period. T_f, T_p and T_g are expressed by the sets of the peak, flat, and valley time in 24 h respectively, as shown in Eq. (26):

$$\begin{aligned} T_f &= 24T + t_f \\ T_p &= 24T + t_p \\ T_g &= 24T + t_g \end{aligned} \tag{26}$$

t_f is the set of peak time in a day, t_f={9 10 11 16 17 18 19 20 21 22}; t_p is the set of flat time in a day, t_p={8 12 13 14 15 23}; t_g is the set of valley time in a day, t_g={24 1 2 3 4 5 6 7}.T is the sequence from day 1 to day 365.

The environmental benefit should be taken seriously, so the total CO_2emission can be expressed by Eq. (27):

$$CO_{2,total} = \lambda_e \sum P_e(t).\Delta t + \lambda_f \sum P_f(t).\Delta t \tag{27}$$

λ_e is CO_2 the emission intensity of electrical equipment (λ_e=0.02072kg $CO_{2,eq}/kW{\cdot}h$), and λ_f is CO_2 the emission intensity of the gas boiler (λ_f =0.17644kg $CO_{2,eq}/KWh$).

2.4. Bilevel optimization methods

Fig. 3 is the flowchart based on optimization approach for building cold and heat source system optimization of this paper. The system parameters (energy price, equipment efficiency, price of unit capacity) and meteorological data (such as outdoor temperature, solar radiation) are input variables. Based on the DEST simulation prediction, the 8760 h cooling and heating load, building lighting and other electricity load of the building are obtained, which are act as input variables into the mixed integer nonlinear optimization model for the planning and operation of the building cooling and heating source system, The planning and operation of the building cooling and heating source system proposed in this paper belongs to a two-level planning model. The upper level is an investment decision problem, with the goal of minimizing the total life cycle cost, including investment cost and annual operating cost (C_{init}, C_{op}), and the variables to be optimized are the capacities of all equipment($Q_{c,ECH,max}$,$Q_{h,FB,max}$,$Q_{h,EH,max}$,V_{TES}^{max},V_{CES}^{max}); the lower level is the operation optimization problem, the goal is to minimize the annual operating cost (C_{op}, $CO_{2,total}$), including fuel cost and carbon emissions, and the optimized variable is the annual operating power of 8760 h of different equipment ($P_f(t)$, $P_{e,ECH}(t)$, $P_{e,EH}(t)$ *etc.*). The upper-level

Table 1
Time-of-use tariff in Changsha.

Period	Peak	Flat	Valley
Periods	8:00–11:00, 15:00–22:00	7:00–8:00, 11:00–15:00, 22:00–23:00	23:00–7:00
Price	1.0995 RMB/kW•h	0.7003 RMB/kW•h	0.2151 RMB/ kW•h

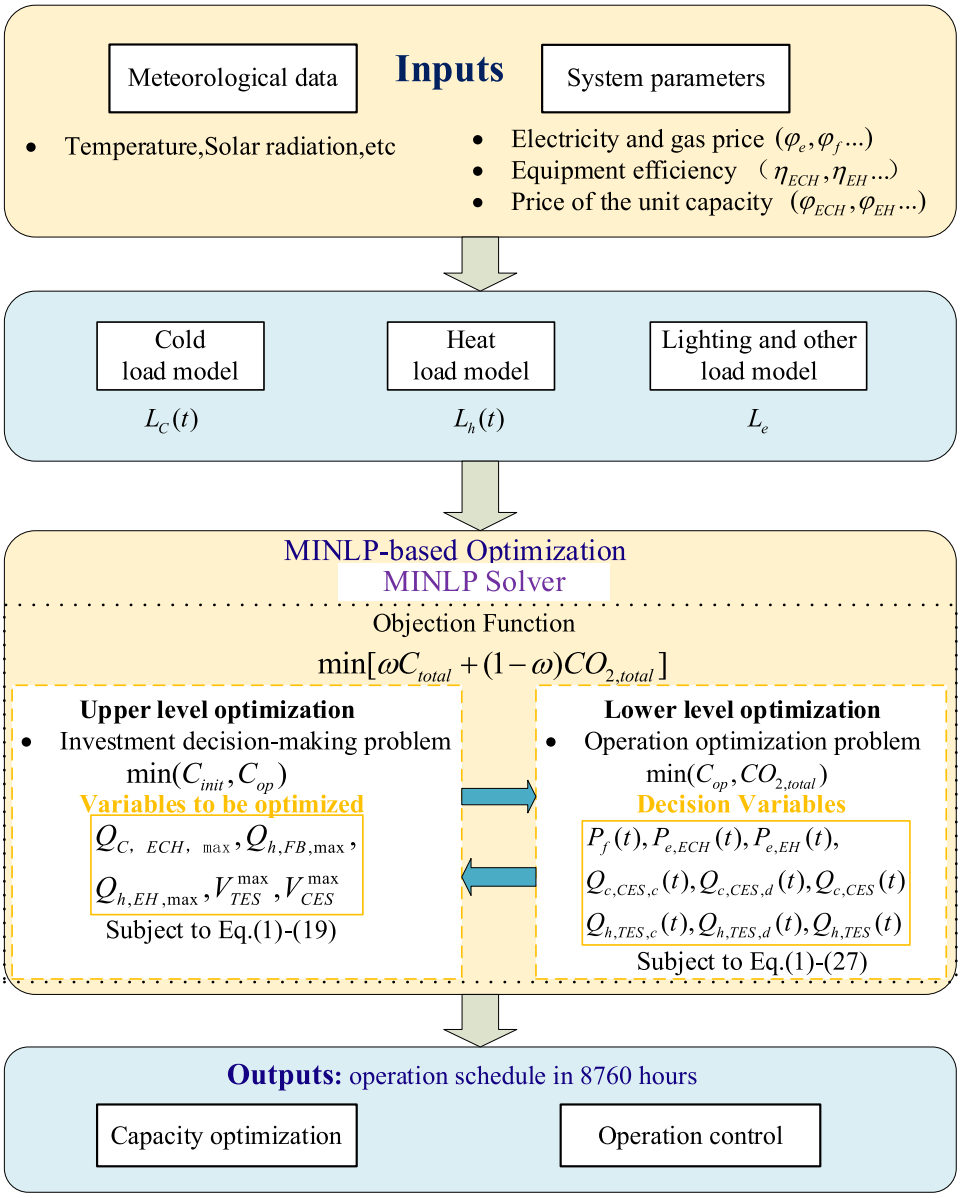


Fig. 3. The flowchart based on our optimization approach for cooling and heating source system.

investment decision result acts on the lower-level objective function and constraint conditions, and the lower-level operation optimization feeds back the optimal value to the upper-level to realize the interaction between the upper and lower levels. The upper-level planning model constraints involve energy balance and equipment operation constraints (Eq. (1)-(19)), and the lower-level planning model involves energy balance, environmental constraints, equipment operation constraints, economic constraints, etc., Eqs. (1)-(27). Through the MINLP solver, the upper and lower layers mutually restrict iterative calculations, and finally output the 8760 h annual operation schedule, obtain the optimal capacity configuration and operation control, and realize the planning and operation optimization of the building's cold and heat source system.

3. Results and discussion

The above established cold and heat source system model in previous part is solved and optimized, this part mainly studies and discusses the applicability of two types of cold and heat source system in building energy flexibility and grid friendliness.

3.1. Capacity configuration optimization in different weights

Operational capacities optimization is the core of planning and operating of building energy systems. Its purpose is to determine the optimal operation configuration of energy equipment, and to achieve energy matching and energy management between the supply side and demand side.

Fig. 4 illustrates the changes of equipment optimal operating capacities under different weights. As can be found from Fig. 4, the capacity of gas boiler and electricity chiller in case I does not change with ω weights, and their values are 217.24 kW and 803.24 kW, respectively. Compared with case 1, when ω weights are 0.05–1, the capacity of the electricity chiller is 510.63 kW in case II, and the use of the energy storage system reduces the capacity of the electricity chiller by 36.4%. For gas boiler, when ω 0.05, with the increase of ω weights, the capacity of gas boiler increases gradually, and the maximum capacity of gas boiler is 128.3 kW. Compared with the 271.24 kW of case I, the use of energy storage system reduces the capacity of gas boiler by 52.7%. It is worth noting that in the range of 0–0.05, the objective function is to minimize carbon emissions and do not use gas boiler, therefore, the

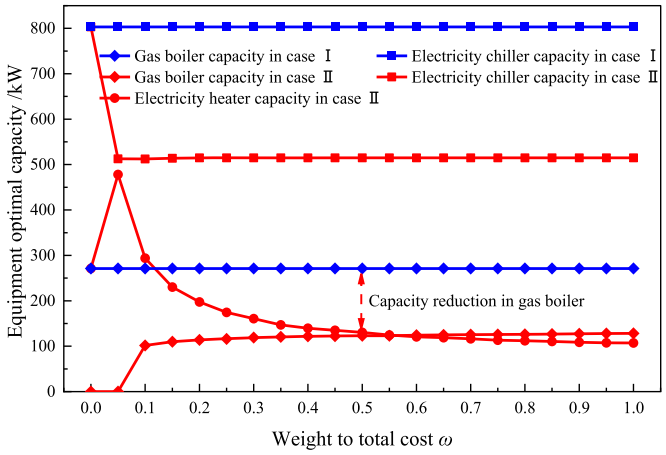


Fig. 4. The equipment optimal capacities under different ω scenarios.

capacity of gas boiler is 0.

Fig. 5 is the changes of cold and thermal storage tank under different ω scenarios. As can be seen from Fig. 5, the volume of the thermal storage tank and cold storage tank is 0 when ω= 0, because the objective function is to minimize carbon emissions, and there is unnecessary to consider the use of gas boiler and energy storage tanks. When ω> 0, the volume of the heat storage tank decreases gradually with the increase of ω weights. Compared with the heat storage tank, the volume of cold energy storage tank rarely changes with the variation of ω weights.

Fig. 6 shows the changes of electricity and gas power consumption of the building energy system all around year with weight ω in the dual-objective optimization. It can be found from the figure that the electricity consumption and gas consumption in case I (i.e. electricity chiller plus gas boiler for cooling and heating demand) do not change under different ω scenarios, and the TOU (time-of-use tariff) has no influence on the energy utilization strategy of the energy-supply system without energy storage. While as the introduction of the energy storage systems, with the increase of weight ω, the electricity consumption in energy storage system (case II) gradually decreases, in comparison with case I, i.e. electricity chiller plus gas boiler for cooling and heating demand, the electricity consumption in case II is greater, the excess proportion in electricity consumption mainly employed to store thermal and cold energy in valley electricity price time in response to the peak and valley electricity price difference. With regard to the natural gas, as ω increases, the amount of gas consumption with energy storage system increases. The gas consumption of case I without energy storage does not change, while the gas consumption in case II with energy storage system

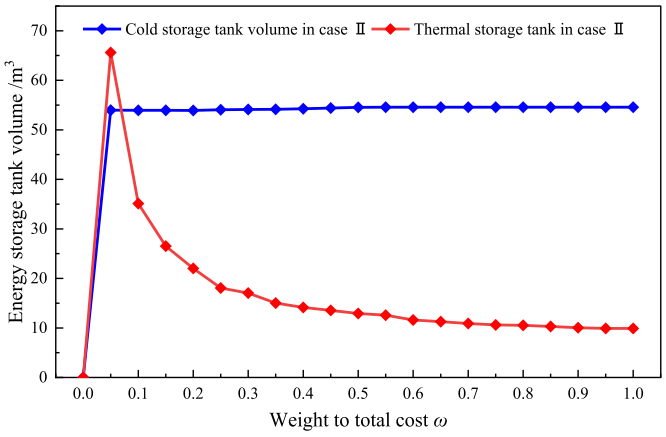


Fig. 5. The changes of cold and thermal storage tank under different ω scenarios.

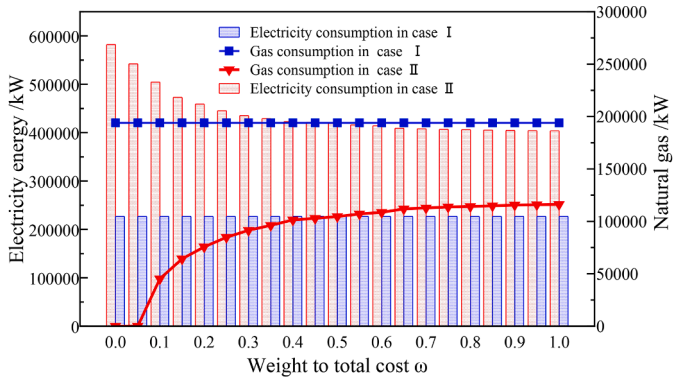


Fig. 6. Building energy consumption under different ω scenarios.

is much smaller than case I without energy storage.

Fig. 7 presents the relationship of total cost during the entire life cycle and total carbon emission of the building energy system. The total cost and carbon emission of case I without energy storage both are constant under different ω scenarios. The reason is that the conventional energy-supply system (case I) only equipped with gas boiler and electricity chiller to provide cooling and heating energy for the building. Under the time-of-use electricity price, there is no alternative form of cooling or heating replacement and electricity chiller and gas-boiler cannot respond to changes of electricity prices. The building energy consumption completely conforms with air conditioning load in conventional energy-supplying system, and the building load cannot be adjusted.

On the contrary, the total carbon emission is positive proportionate to the weight ω in case II with energy storage and the scope is greater in quantity, when the weight coefficient is greater than 0.75, carbon emission is no longer significantly change. By the comparative analysis, the carbon emission of case II with energy storage system proposed in this paper is far lower than the heating and cooling scheme in case I without energy storage. With the increase of weight ω, the total cost of the energy storage system gradually decreases, when $\omega > 0.5$ the total cost reduction tendency is flat and no obvious change (Fig. 7 in 0.25–0.5 range, total cost reduction difference is nearly 3000 RMB each other). The reason is when weight ω at 0–0.5, the priority of the objective function to reduce carbon emissions, so the utilization of electricity energy should be encouraged, and lead to the increase of total cost . When $\omega > 0.5$ the optimal operation capacity of each energy equipment almost is nearly equivalent under different ω scenarios (as found in Fig. 4), the initial investment of each weight ω are approximate. The reduction of electricity power operation cost as well as the accumulation of gas operation cost simultaneously, which causes the total cost of case II with energy storage does not change when $\omega > 0.5$ (depicted in Fig. 7).

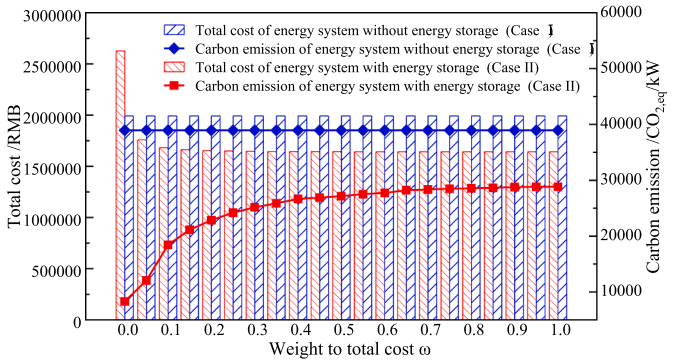


Fig. 7. Total cost and carbon emission of two types of cold and heat source systems under different ω scenarios.

The three situations of $\omega = 0, \omega = 0.5, \omega = 1$ are usually discussed in dual-objective optimization. So Tables 2 and 3 respectively present the optimal operation capacity and cost-effectiveness and environmental results of the established model. According to the optimization results, when $\omega = 0$, the operation capacity of electricity chiller both case II and case I are 803.24 kW. The reason is that the optimization target of energy storage system only for carbon reduction, regardless of the cost reduction, the utilization of cold energy storage tank is redundant, the global cooling requirements completely provided by electricity chiller, the electricity operation power is proportional to the cold load, the cold load is all same, therefore the operation capacity of electricity chiller both are equivalent. Regarding the operation capacity of electricity heater, with the increase of weightω its operation capacity decreases. In order to maintain the balance of power grid and energy flexibility, the operation capacity of electric heater should be reduced. Taking a comprehensive consideration, the weight 0.5 scenario (both minimize total cost and carbon emission) is selected as optimized designing scheme. Under this scheme, the optimal operation capacity of the electricity chiller and electricity heater will be reduced, and even when the energy equipment run with peak power, the adverse impact on the power grid will be weakened. As can be found from Table 3, the total cost and carbon emission of case II with energy storage system are 1,573,686 RMB and 26,353 $CO_{2,eq}$ under 0.5 scenario, and the case I (non-energy storage system) are 1,885,090 RMB and 38,116 $CO_{2,eq}$ respectively. The total cost can be saved by 16.5% and carbon emission can be reduced by 30.9%.

3.2. Estimation of energy-flexible and grid-friendly performance

Air-conditioning energy consumption accounts for a large proportion of building energy consumption, and in most situations, the time of peak power of electricity chiller coincides with the peak time of power-grid when electricity chiller unit is running. With regard to commercial buildings and office buildings, the load coefficient of electricity chiller can be used to characterize the power profile of power grid. Therefore, this paper introduces load coefficient LC to evaluate the flexibility performance of building energy system and reflect the friendliness of power grid. The load coefficient is defined as the ratio of the average load to the peak load in Eq. (28) during the specific time [25].

$$LC = \frac{P_{average}}{P_{peak}} \tag{28}$$

From the definition, a high LC reflects the smaller difference between the building air-conditioning peak power and the average power consumption, which indicates that the building air-conditioning power profile is more stable.

Fig. 8 presents the power profile of electricity chiller with energy storage system and without energy storage when the optimal operation capacities are determined. It can be obviously found from the figure that the operating power of the electricity chiller in case I without energy storage complies with the cold load. The peak chiller power and the max cold load appear simultaneous. During the cooling season (May-Aug), the operating power of electricity chiller is greater and reaches a peak power of more than 200 kW in hot summer, and the load coefficient of electricity chiller of case I without energy storage is 40.9%. Regarding case II with energy storage, the operating power of electricity chiller is relatively stable throughout the year, and electricity chiller works with full load as well as the load coefficient of electricity chiller is 83.34%. By comparison, it is found that the cold load of electricity chiller in case II with energy storage during the daytime is shifted into nighttime, and the power profile of the electricity chiller unit is flatten and stable. The flatten curve indicates a better load profile and a higher load coefficient. On the contrary, the power distribution curve of the electricity chiller in case I is relatively steep and random. In hot summer, the operating power of electricity chiller is greater than case II with energy storage, and the load coefficient is low, which is more dependent on the power grid and less energy-flexible performance. On the other hand, the case II with energy storage reduces the work hours of the electricity chiller during daytime and promotes the reduction of capacity of the power-grid network and strengthens the friendliness of the grid.

Table 2
The optimized results of capacities confirmation.

	Case II with energy storage system(ω=0)	Case II with energy storage system (ω=0.5)	Case II with energy storage system (ω=1.0)	Case I without energy storage system (ω=0,0.5,1.0)
$Q_{c,ECH,max}$	803.24kW	510.63kW	512.4kW	803.24kW
$Q_{h,EH,max}$	271.24kW	130.27kW	107kW	0
$Q_{h,FB,max}$	0	123kW	128.3kW	271.24kW
V_{CES}^{max}	0	54.5m^3	54.5m^3	0
V_{TES}^{max}	0	13m^3	9.9m^3	0

Table 3
The economic-environmental benefits.

	Case II with energy storage system(ω=0)	Case II with energy storage system(ω=0.5)	Case II with energy storage system(ω=1.0)	Case I without energy storage system (ω=0,0.5,1.0)
C_{init}	338043RMB	248268RMB	244537RMB	330177RMB
C_{op}	2171356RMB	1325418RMB	1328223RMB	1554913RMB
C_{total}	2509400RMB	1573686RMB	1572760RMB	1885090RMB
$CO_{2,total}$	7956.04$CO_{2,eq}$	26,353$CO_{2,eq}$	27,999$CO_{2,eq}$	38,116$CO_{2,eq}$

Fig. 9 indicates the power profile of gas boiler in two types of cold and heat source systems. The load coefficient is adopted to characterize the friendliness of heating network and power grid. The operating power of gas boiler in conventional system (case I without energy storage) is subjected to heat load. The peak power of gas boiler and the maximum heat load found simultaneously, and the power curve is steep. The load coefficient of gas boiler is 29.7%. Regarding to case II with energy storage, due to the adoption of electricity heater, the power profile of gas boilers changes and the curve becomes flatten, with a load coefficient of 41.65%, and the load coefficient of electricity heater is also above 30.65%. The high load coefficient of electricity heater and gas boiler is favorable to the balance of power grid and the pressure-reduction of natural gas pipelines, as well as improves the operation stability and reliability of the building energy system.

3.3. Load demand management control in typical day

In general, the electricity price at valley time is cheaper than peak time, therefore the electricity consumption at peak time can be transferred to the valley time to perform building energy flexibility. Fig. 10 reveals the relationship of stored cooling energy with time in summer typical day. A considerable part of cold capacity is generated during the valley electricity price at 0–7 h in the morning and stored in cold energy storage tank in the form of chilled water (i.e. charging cold period). During the daytime when electricity prices are peak, the stored cold energy is released to handle the cooling load of the building, which can greatly reduce the electricity power demand during peak hours.

Fig. 11 shows the cooling load distribution in typical summer day. It can be seen from the figure that in the case II with energy storage system, the electricity chiller works at full load from 19:00 to 22:00 to provide part-load demand and store cold energy. When the electricity price is high during daytime, turn chiller on when the stored cold energy is not enough to compensate for cold load. The building cooling load reaches the first peak at 9:00, and the cold storage water tank is employed to compensate the load demand. After the cold storage water tank is used, at the second peak of the cooling load (21:00), only a part of fixed

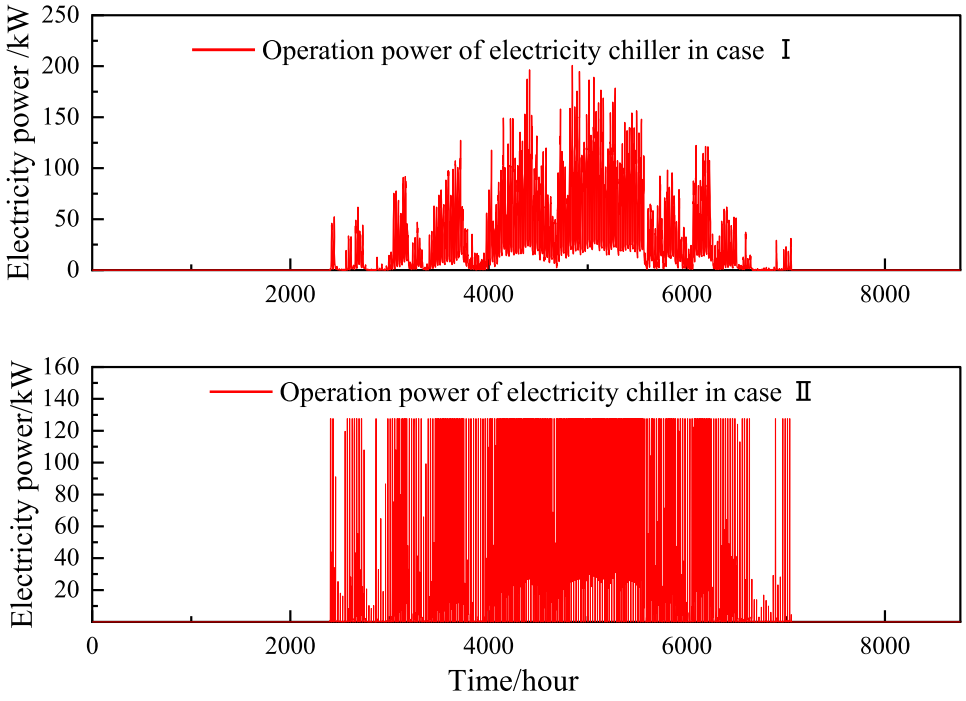


Fig. 8. Power profile of electricity chiller around year in two types of cold and heat source systems.

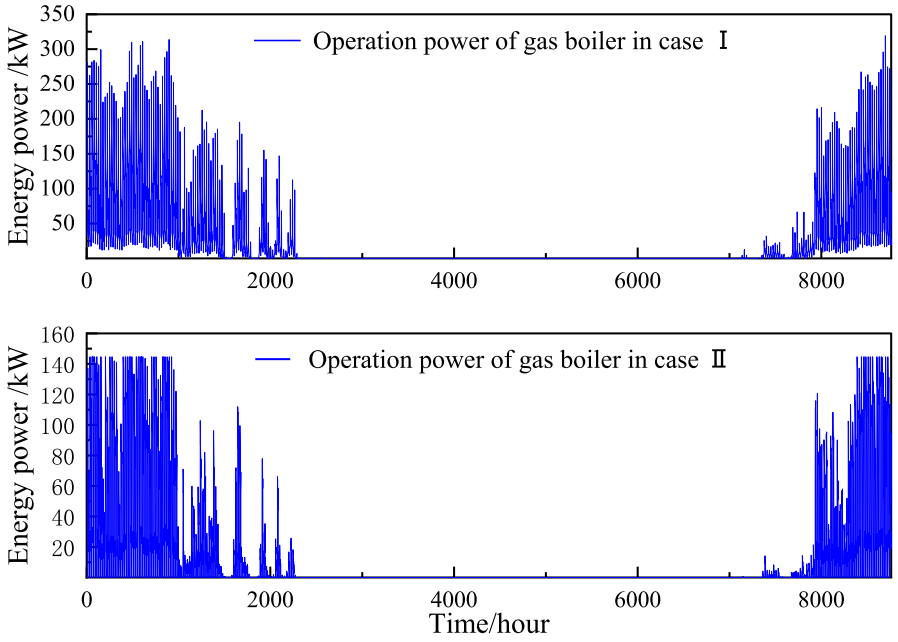


Fig. 9. Power profile of heating system in two types of cold and heat source systems.

cooling load is borne by electricity chiller, and the rest of cold load is met by cold storage tank. Because the adoption of cold storage system, the load at peak moment is transferred to valley time, and the power profile of refrigeration unit is relatively stable. This flatten power distribution improves the load factor of electricity chiller, reduces the grid capacity during the peak building load period, which is favorable to address the grid long-term power imbalance problem and improves energy flexibility. On the contrary, regarding the conventional system case I without energy storage, chiller usually dominates the power demand of the building, which causes more serious problems such as the power of chiller almost proportional to cold load, the load factor is lower as well as the dependence on power grid is stronger. Furthermore, the grid capacity increases with the operating capacity of electricity chiller increases, and the power-supply frequency is unstable, which damages the healthy operation of power grid and has less energy flexibility.

Fig. 12 shows the heating load distribution in typical winter day. Regarding case II with energy storage system, a large part of heat load is shifted to nighttime, and electricity heater makes full advantage of the electricity energy to produce thermal energy during the valley price period and store the heat by the thermal storage tank. The stored heat energy is discharging when heat load is greater (9:00–14:00); when the peak heat load occurs, the thermal storage tank and gas boiler provide

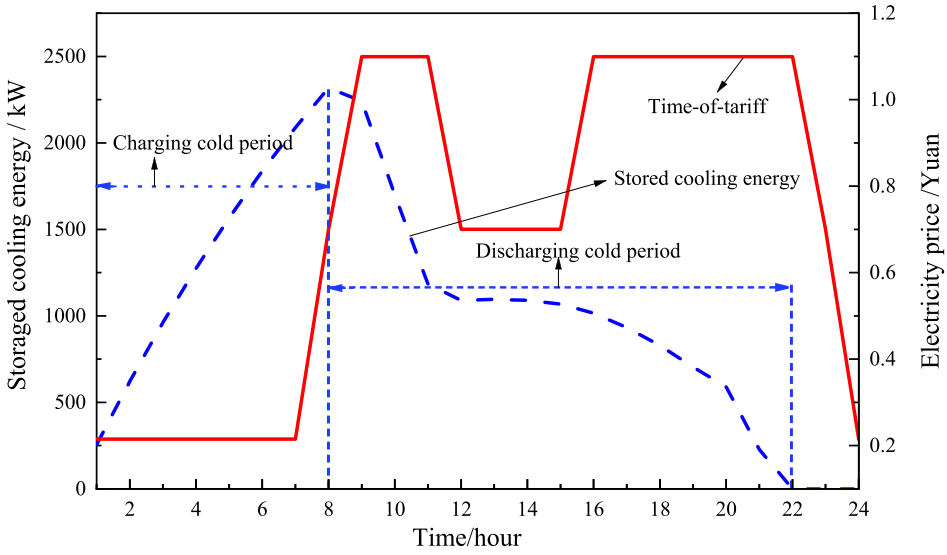


Fig. 10. The stored cooling energy with time-of-use electricity price.

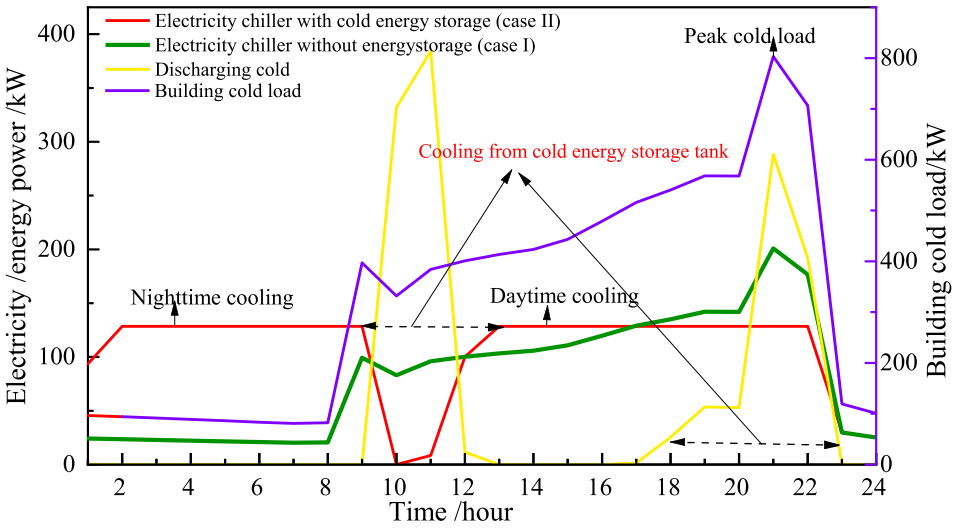


Fig. 11. The cooling load distribution in typical summer day.

simultaneously heating energy. After 14:00, the electricity price is higher, and only the gas boiler as heating source meets thermal load. The adoption of thermal energy storage tank improves the load coefficient of gas boiler and electricity heater, as well as reduces the operating capacity of gas boiler and electricity heater, which improves the stability of power grid and building electricity energy flexibility. On the contrary, the operating power of gas boiler in conventional system case I is directly proportional to thermal load, which leads to the large operating capacity and low load factor and worsens system stability.

In order to verify the availability of the selected (ω=0.5) scenario, three energy flexibility scenarios (ω=0,ω=0.5,ω=1) are discussed and presented in Fig. 13. It can be seen from Fig. 12 that when cost-effectiveness is not considered (ω=0), the operating power of the equipment is completely determined by building load, and the power of electricity heater is proportional to building heat load. When ω=1, the operating capacity of electricity heater is large but the load coefficient is low. The dependence on the power grid is strong, which is susceptible to lead to power-grid imbalance problem and energy inflexibility. The operating capacity of electricity heater has reduced while the operating capacity of gas boiler has increased in weight 1 scenario. Although this scheme enhances the friendliness of power grid, it can be found from Table 3 that the carbon emission is serious. From previous analysis, when both cost-effectiveness and carbon emission are considered simultaneously (ω=0.5), the operating capacity of gas boiler has reduced compared with ω=1 scenario, and the operating capacity of electricity heater is not much different from ω=1, as well as low carbon emission, high load factor, and stable power grid operation. It is noteworthy that under three scenarios, since 14:00–23:00 electricity price is expensive than natural gas, only gas-fired boiler is employed for heating after 14:00, which is beneficial to cost-effectiveness. In general, the scenario of (ω=0.5) has more potentials (such as high load coefficient of energy equipment, cost-effectiveness, environment-friendly performance) to maintain power grid balance as well as building energy flexibility and energy management. It is notable that the energy flexibility provided by energy storage system (case II) is actually realized through the response to the electricity load change of the HVAC system to the electricity price. The supplement of cold and heat storage system on the basis of the cold and heat source scheme with electricity chiller and gas-boiler in the hot summer and cold winter zones is helpful to alleviate the fluctuation of energy demand, maintain the balance of

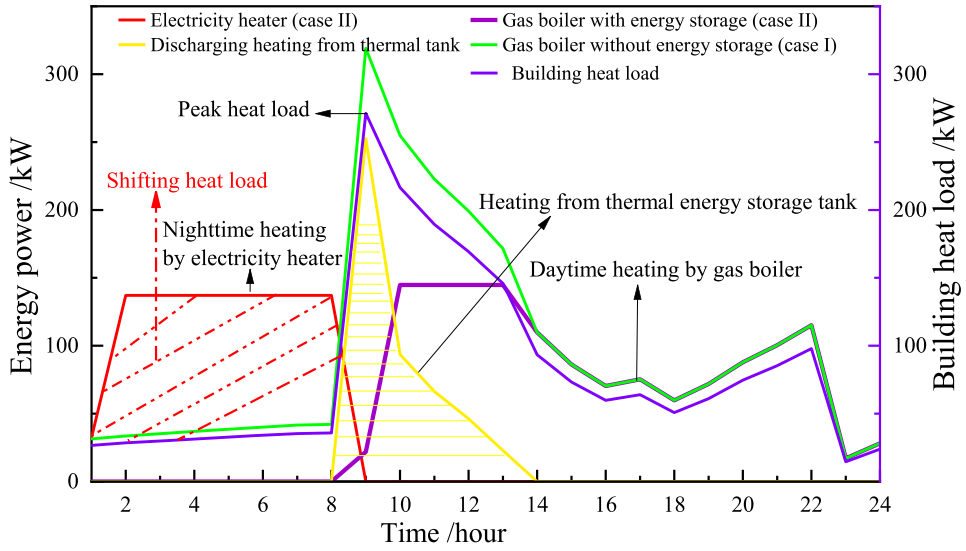


Fig. 12. The heating load distribution in typical winter day.

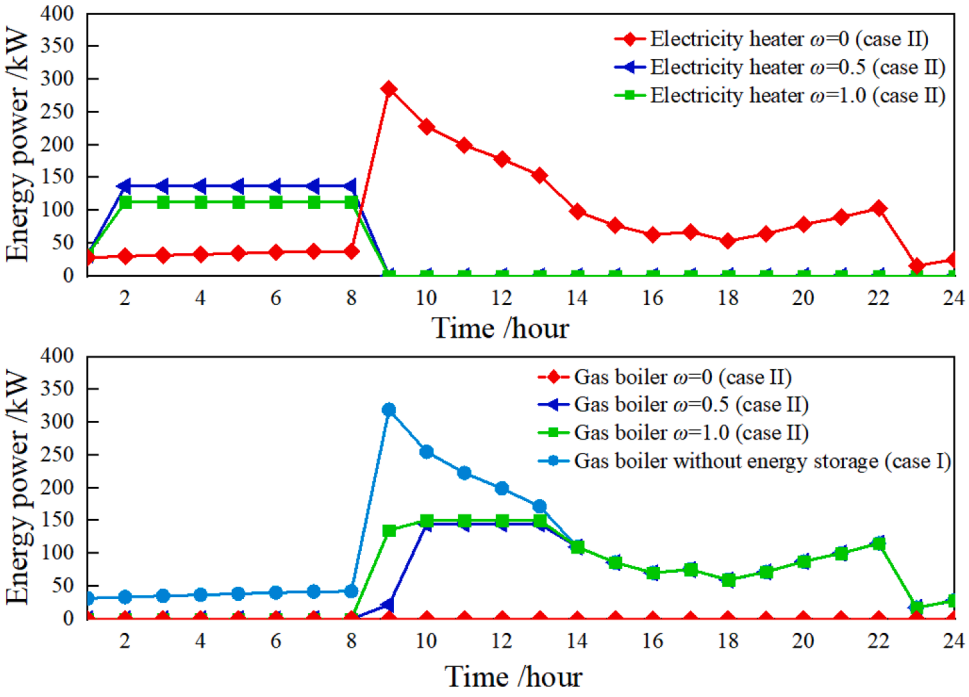


Fig. 13. Building energy flexibility corresponding to different ω sections.

power grid and realize building energy flexibility.

4. Conclusions

In order to realize the building electricity energy flexibility in hot summer and cold winter areas, this paper improved the potential of building energy flexibility by adding energy storage devices on the basis of the traditional cold and heat source scheme (case I). In this paper, a multi-objective optimization model of energy system is established and solved, and the capacity configuration and operation optimization of the system is optimized based on bilevel optimization methods, and the performances of building energy flexibility of case II are analyzed and discussed. The main conclusions are as follows:

(1) Taking both the total cost and carbon emission into consideration (when ω = 0.5), compared with case I, the optimal operation capacity of gas-boiler and electricity chiller in case II can reduce to 123 kW and 510.63 kW respectively, and the total cost in the whole life period of case II can be saved by 16.5% and carbon emission is reduced by 30.9% when energy equipment capacity configuration is optimal.

(2) The implement of energy flexibility by energy storage system can significantly improve the load coefficient and grid-friendly performance of energy devices. Compared with case I, the load coefficient of electricity chiller in case II is increased from 40.9% to 83.34%, the load coefficient of gas-boiler is increased from 29.7% to 41.6%, and the load coefficient of electricity heater is 30.65%, which represents that supplement of energy storage system in hot

summer and cold winter zones improves building energy flexibility and enhances grid friendliness.

(3) The load demand management control is performed for two types of cold and heat source systems in typical cooling day and heating day, it is found that compared with case I, case II can reduce the peak demand of cold and heat, the power distribution of electricity chiller and electricity heater are stable, and the adaptability of building energy flexibility is better. The energy flexibility provided by case II is actually realized by the response of the change of electricity load of HVAC system to electricity price, the supplement of cold and heat storage tank can effectively mitigate the long-term power imbalance and stabilize the fluctuation of power grid, reduce the dependence of buildings on the power grid.

Credit authorship contribution statement

Jiangyang Liu: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Software, Writing - original draft &editing.

Xu Yang: Software and Validation

Zhongbing Liu: Conceptualization, Methodology, Validation, Investigation, Formal analysis, review, Supervision and Funding acquisition.

Juan Zou: Software and Methodology, Writing - review & editing

Yaling Wu: Software, Formal analysis, review.

Ling Zhang: Writing - review & editing, Funding acquisition

Yelin Zhang:Writing - review & editing

Hui Xiao: Writing - review & editing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Nomenclature

Abbreviations

GB Gas boiler
ECH Electricity chiller
EH Electricity heater
CES Cold storage water tank
TES Thermal storage tank
TOU Time-of-use tariff
MINLP Mixed integer nonlinear programming

Parameters

η_{TES} Heat storage efficiency
$\eta_{TES,d}$ Heat discharging efficiency
$\eta_{TES,c}$ Heat charging efficiency
η_{CES} Cold storage efficiency
$\eta_{CES,d}$ Cold discharging efficiency
$\eta_{CES,c}$ Cold charging efficiency
η_{ECH} Cooling efficiency of ECH
η_f Heating efficiency of GB
η_{EH} Heating efficiency of EH
K_1 Safety factor considering energy loss
K_2 Factor of energy delivery loss
ρ Water density
c_p Specific heat of water
T_{max} Average maximum temperature
T_{min} Average minimum temperature
H Energy storage time in valley period
i Annual interest rate
n Payback period
φ_{FB} Price of the unit capacity of GB
φ_{EH} Price of the unit capacity of EH
φ_{CH} Price of the unit capacity of ECH

Parameters

φ_f Unit price of natural gas
λ_e Emission intensity of electrical equipment
λ_f Emission intensity of the natural gas
φ_{TES} Price of the unit volume of TES
φ_{CES} Price of the unit volume of CES
L_e Lighting and other electricity demand

Variables

$P_{e,ECH}(t)$ Power consumption of ECH
$P_{e,EH}(t)$ Power consumption of EH
$P_f(t)$ Natural gas power consumed by GB
$P_e(t)$ Total electricity power demand
$Q_{h,EH}(t)$ Heating capacity of EH

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(*continued*)

$Q_{h,FB}(t)$ Heating capacity of FB
$Q_{h,TES,d}(t)$ Discharging heat energy by TES
$Q_{h,TES,c}(t)$ Charging heat energy by TES
$L_h(t)$ Building heat load
$Q_{c,ECH}(t)$ Cooling capacity of ECH
$Q_{c,CES,d}(t)$ Discharging cold energy by CES
$Q_{c,CES,c}(t)$ Charging cold energy by CES
$L_c(t)$ Building cold load
$Q_{h,TES}(t)$ Net stored heat energy by TES
$Q_{c,CES}(t)$ Net stored cold energy by TES
$Q_{c,ECH,max}$ Optimum cooling capacity of ECH
$Q_{h,FB,max}$ Optimum heating capacity of FB
$Q_{h,EH,max}$ Optimum heating capacity of EH
$Q_{h,TES}^{max}$ Optimum heat storage capacity of TES
$Q_{c,CES}^{max}$ Optimum cold storage capacity of CES
$V_{yES}(t)$ Volume of CES or TES
ω The weight given to the minimum total cost
C_{total} The minimum total cost
C_{init} Initial capital cost
C_{op} Operational cost of by gas or electricity
Variables
$CO_{2,total}$ The minimum total CO_2emission
V_{TES}^{max} Optimum volume of TES
V_{CES}^{max} Optimum volume of CES

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